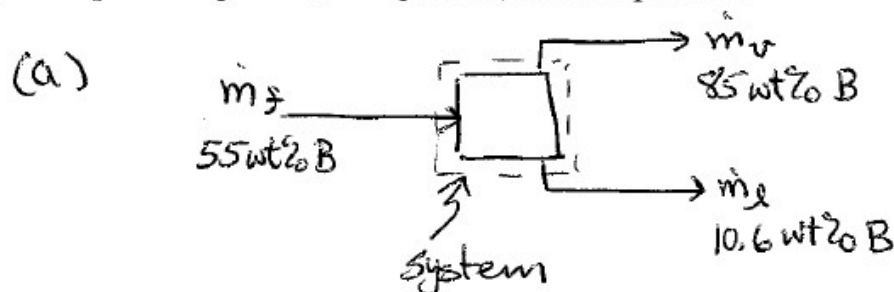


A liquid binary mixture of benzene (C_6H_6) and toluene (C_7H_8) that contains 55 wt% benzene is put into an evaporator to yield a vapor stream that is 85 wt% benzene and a residual liquid stream that is 10.6 wt% benzene.

- (a) Draw a schematic of the process and label all of the streams. The mass flow rate of the feed stream is $\dot{m}_f$ and the output streams are given by $\dot{m}_v$ and $\dot{m}_l$ for the vapor and liquid streams respectively.
- (b) Find the ratio of the mass flow rates of the vapor output to the liquid output ($\dot{m}_v / \dot{m}_l$).
- (c) Is the problem specified, overspecified, or underspecified?



(b) total mass balance: $\dot{m}_f = \dot{m}_v + \dot{m}_l$

benzene mass balance: $0.55 \dot{m}_f = 0.85 \dot{m}_v + 0.106 \dot{m}_l$

There are at least two solution strategies: (1) pick a feed basis, of $\dot{m}_f = 100 \text{ kg/hr}$ for example, and (2) eliminate $\dot{m}_f$ from the two equations and solve for $\dot{m}_v / \dot{m}_l$.

$$\begin{aligned} (1) \quad & \left. \begin{aligned} 100 \text{ kg/hr} &= \dot{m}_v + \dot{m}_l \\ 55 \text{ kg/hr} &= 0.85 \dot{m}_v + 0.106 \dot{m}_l \end{aligned} \right\} \Rightarrow \begin{aligned} \dot{m}_l &= 40.3 \text{ kg/hr} \\ \dot{m}_v &= 59.7 \text{ kg/hr} \end{aligned} \Rightarrow \frac{\dot{m}_v}{\dot{m}_l} = 1.48 \end{aligned}$$

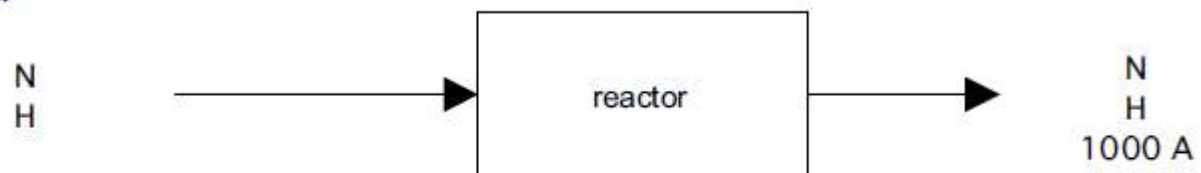
$$(2) \quad 0.55(\dot{m}_v + \dot{m}_l) = 0.85 \dot{m}_v + 0.106 \dot{m}_l$$

$$\frac{\dot{m}_v}{\dot{m}_l} = \frac{0.444}{0.3} = \underline{\underline{1.48}}$$

(c) Since you were only asked for $\dot{m}_v / \dot{m}_l$ and this can be uniquely solved for with the information provide, the problem is specified. Had the problem asked for both $\dot{m}_l$ and $\dot{m}_v$, then it would have been underspecified

P2.39

Diagram:



Components: N (nitrogen as N_2), H (hydrogen as H_2), A (ammonia)

DOF analysis:

	Variables	Constraints
Stream	5	
System	1 (reaction)	
Specified flows		1
Specified compositions		1 (3:1 ratio in feed)
Specified system performance		1 (25% conversion)
Material balance		3

$$\text{DOF} = 6 - 6 = 0$$

Units: ok for now (lbmol/h and mol ratios), will have to convert answer to volumetric flow using ideal gas law.

Basis: 1000 lbmol/h NH_3 in outlet stream.

$$A_{out} = 1000$$

Material balances:

$$N_{in} - N_{cons} = N_{out}$$

$$H_{in} - H_{cons} = H_{out}$$

$$A_{gen} = A_{out}$$

From stoichiometry:

$$N_{cons}/H_{cons} = 1/3, \quad A_{gen}/N_{cons} = 2/1$$

From inlet stream specifications:

$$N_{in}/H_{in} = 1/3$$

From system performance specification:

$$N_{cons} = 0.25N_{in}$$

These equations are solved to find:

$$A_{gen} = 1000$$

$$N_{cons} = 500$$

$$H_{cons} = 1500$$

$$N_{in} = 2000$$

$$H_{in} = 6000$$

$$N_{out} = 1500$$

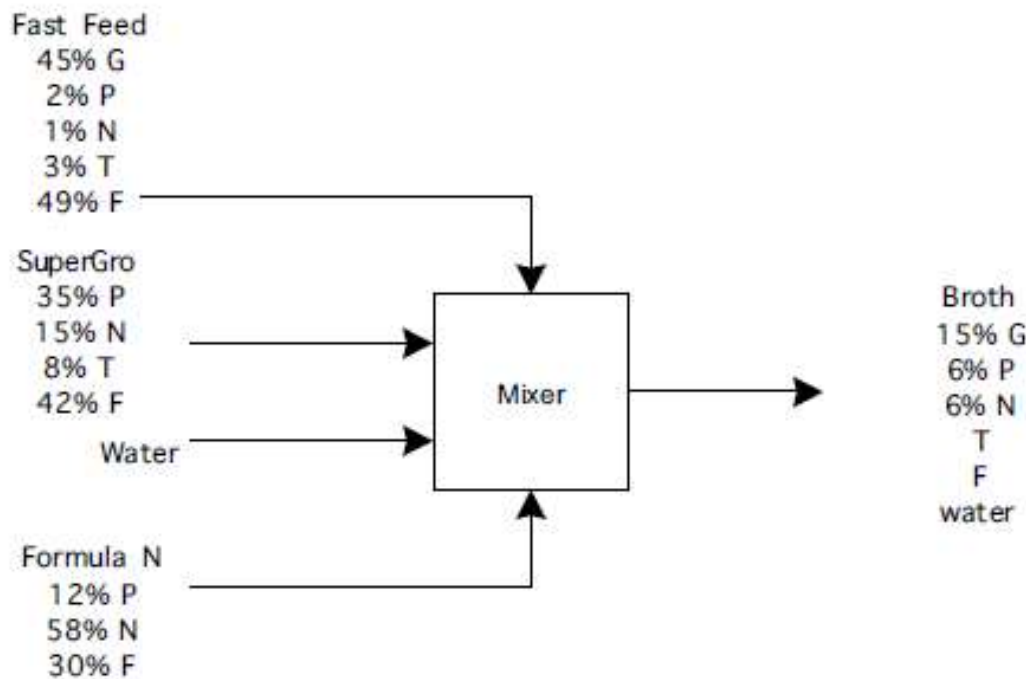
$$H_{out} = 4500$$

The volumetric feed rate to the reactor is calculated from the inlet molar flows and the ideal gas law:

$$(8000 \text{ lbmol/h}) \left(\frac{0.7302 \text{ atm ft}^3}{\text{lbmol } ^\circ\text{R}} \right) \left(\frac{1440 \text{ } ^\circ\text{R}}{100 \text{ atm}} \right) = 8.4 \times 10^4 \text{ ft}^3/\text{h}$$

P2.58

The flow diagram, with the mixer as system, is



From a DOF analysis, we find that there are 19 stream variables, 12 specified stream compositions, 1 specified basis (1 kg broth), and 6 material balances. $DOF = 19 - (12+1+6) = 0$.

(a) Since we are given the compositions, but not the quantities, of each of the powders that go into the mix, it is easier to write the stream variables as the (known) mass fraction times the (unknown) total quantity of the powder. We will use “FF”, “SG”, and “FN” to indicate the quantities of the 3 powders. The material balance equations are:

Glucose: $0.45FF = 0.15(1) = 0.15 \text{ kg}$, so $FF = 0.333 \text{ kg}$

Phosphate: $0.02FF + 0.35SG + 0.12FN = 0.06(1) = 0.06 \text{ kg}$

Nitrate: $0.01FF + 0.15SG + 0.58FN = 0.06(1) = 0.06 \text{ kg}$

Substituting in the known value for FF and solving simultaneously, we find

$SG = 0.13 \text{ kg}$, and $FN = 0.064 \text{ kg}$

Each kg of broth should contain 0.333 kg Fast-Feed, 0.13 kg Super-Gro, and 0.064 kg Formula N, plus 0.473 kg water.

(b) The cost per kg of broth (excluding any operating costs such as labor or electricity) made from these ingredients is

$$(0.333 \times \$20) + (0.13 \times \$10) + (0.064 \times \$15) = \$8.92$$

Since the purchased broth costs \$15/kg, we can save \$6.08/kg (at most) by mixing our own. With a 20 kg batch, we save $(20 \times \$6.08)$ or \$121/batch. If the equipment costs \$2000, we need to make up $(\$2000/\$121/\text{batch})$, or 16 batches total, to pay for the equipment.

(c) The kg trace nutrients in the broth made according to the recipe in part (a) is

$$0.03(0.333) + 0.08(0.13) = 0.020 \text{ kg}$$

or, the wt% is 2.0. Requiring 3 wt% trace nutrients adds an additional constraint, (DOF = -1), and the problem is overspecified. We cannot make a broth that exactly matches all constraints.

To provide 3 wt% trace nutrients, we need to increase Super-Gro (it is cheaper than Fast Feed, and is the only powder with >3% trace nutrients).

$$0.03(0.333) + 0.08SG = 0.030 \text{ kg, so } SG = 0.25 \text{ kg.}$$

We cannot decrease the quantity of Fast-Feed, because that is our only source of glucose. We can, however, decrease the amount of Formula-N, which is attractive because it is more expensive than Super-Gro. From the balances on nitrates, we find that we can reduce Formula N to 0.033 kg. With this, the final mix would contain 15wt% glucose, 6 wt% nitrates, and 3 wt% trace nutrients, per the specifications, and would exceed phosphate requirement (9.8 wt%, vs 6 wt% required.) The quantity of water would decrease to 0.384 kg, and the price would increase to

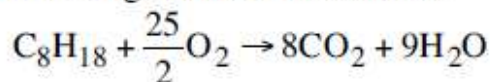
$$(0.333 \times \$20) + (0.25 \times \$10) + (0.033 \times \$15) = \$9.66.$$

P2.60

If one-quarter of all CO₂ emissions are attributed to US, then the pounds of CO₂ produced per person in US is

$$\frac{\frac{0.25 \times 5.6 \times 10^9 \text{ metric tons}}{\text{yr}} \times \frac{2200 \text{ lb}}{\text{metric ton}}}{300 \times 10^6 \text{ persons}} = 10,266 \text{ lb CO}_2/\text{person}$$

Combustion of gasoline is modeled as



From this we calculate CO₂ emissions per g of gasoline:

$$\frac{8 \text{ gmol CO}_2}{\text{gmol C}_8\text{H}_{18}} \times \frac{44 \text{ g/gmol}}{114 \text{ g/gmol}} = \frac{3.1 \text{ g CO}_2}{\text{g C}_8\text{H}_{18}}$$

If average person drives 20 miles/day (7300 mi/year) and gets 27.5 miles/gallon, with gasoline at a specific gravity of 0.7, then each person is responsible for

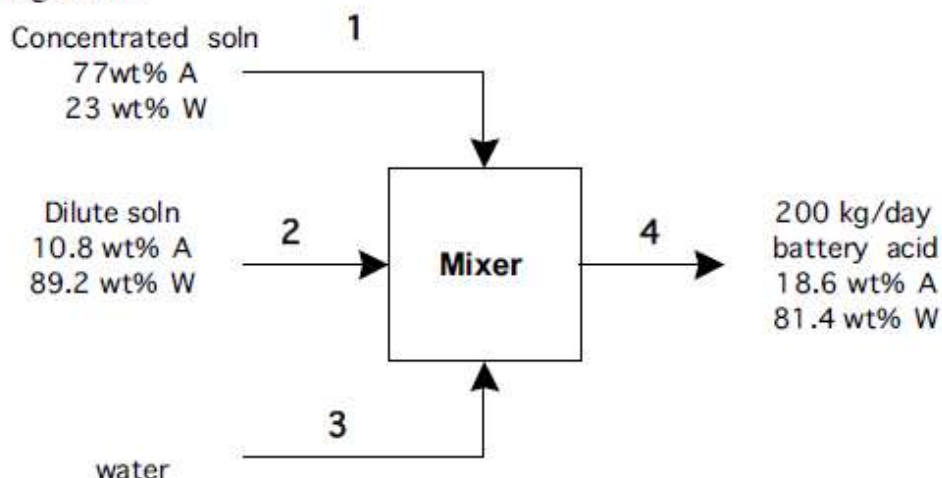
$$\frac{7300 \text{ mi}}{\text{y}} \times \frac{1 \text{ gal}}{27.5 \text{ mi}} \times \frac{1000 \text{ mL}}{0.26417 \text{ gal}} \times \frac{0.7 \text{ g C}_8\text{H}_{18}}{\text{mL}} \times \frac{3.1 \text{ g CO}_2}{\text{g C}_8\text{H}_{18}} \times \frac{1 \text{ lb}}{454 \text{ g}} = 4800 \text{ lb CO}_2$$

(Estimates for average fleet mileage, miles driven per vehicle, and miles driven per person are somewhat variable depending on source; numbers seem to hover around 25 miles/gallon, about 15,000 miles/vehicle, and about 7000-8000 miles per person.)

This is just under half of the lb CO₂/person calculated above. The estimate from the environmental group seems quite reasonable, because we have accounted for CO₂ generation only from automobiles, and not from electricity, home heating, public transportation, manufacturing, etc.

P2.64

The flow diagram is:



Components are sulfuric acid (A) and water (W).

DOF analysis:

	Variables	Constraints
Stream	7	
System	0	
Specified flows		1
Specified compositions		3 (% acid in streams 1, 2 and 4)
Specified system performance		0
Material balance		2

$$\text{DOF} = 7 - 6 = 1$$

The problem is underspecified. This is an opportunity for optimization – we want the “best” combination of flows into the mixer, with the best being defined as “minimum cost”.

Units are all in mass/day or wt%, no conversions needed.

With the compositions known and the flow rates unknown, it is easiest to work with variables defined as mass fractions and total mass flows, rather than mass flows of individual components.

Material balances:

$$\text{Acid: } 0.77F_1 + 0.108F_2 = 0.186(200) = 37.2 \text{ kg/day}$$

$$\text{Water: } 0.23F_1 + 0.892F_2 + F_3 = 0.814(200) = 162.8 \text{ kg/day}$$

There are 2 equations in 3 unknowns. We can add another equation (in another unknown) to describe the cost of the raw materials:

$$0.10F_1 + 0.02F_2 = \text{Cost}(\$/\text{day})$$

There are a couple of ways to solve this problem.

Method 1: Brute force: We set one of the unknowns (e.g., F_1) at a value between 0 and 200 kg/day, then calculate the flows of the remaining streams and the total cost. By varying F_1 , we can then find the cost as a function of F_1 , and pick the value of F_1 that minimizes the cost. These calculations are easily carried out in a spreadsheet. From these calculations, we first note that there is only a narrow range of possible values of F_1 , where F_2 and F_3 are also between 0-200 kg/day, from $F_1 = 23.6$ to $F_1 = 48.3$ kg/day. If we plot the cost vs F_1 , we note that the cost decreases linearly with increasing F_1 , so the minimum cost (within the other constraints) is achieved with $F_1 = 48.3$ kg/day (and $F_2 = 0$, $F_3 = 151.7$ kg/day).

Method 2: Thinking: we note that per kg of sulfuric acid, we are paying $(\$0.10/(0.77))$ or $\$0.13/\text{kg}$ sulfuric acid in the concentrated form, and $(\$0.02/0.108)$ or $\$0.185/\text{kg}$ sulfuric acid in the dilute form. Obviously, we want to use as much of the “cheaper” sulfuric acid as possible, so we want to use as much of the concentrated solution as possible. We want to set F_2 as low as possible – indeed, we can set $F_2 = 0$ and satisfy all constraints. We get the same answer as in Method 1: the minimum cost is obtained with $F_1 = 48.3$ kg/day, $F_2 = 0$, $F_3 = 151.7$ kg/day, and the cost is $\$4.83/\text{day}$.

